IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Intelligent Classification of Rural Infrastructure Projects

Domain of Project – Government / e-Governance

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Aditya Kumar		adityakumar.innet@gmail.com	

Problem Statement -Intelligent Classification Of Rural Infrastructure Projects

The Challenge

The Pradhan Mantri Gram Sadak Yojana (PMGSY) includes multiple schemes (PMGSY-I, PMGSY-II, RCPLWEA, etc.), making project classification and management complex. Manual classification is time-consuming, error-prone, and affects project monitoring, fund allocation, and decision-making.

The Objective

Develop an AI-powered PMGSY Scheme Classification System that:

1. **Automatic Scheme Classification** – Predicts the correct PMGSY scheme from project attributes.
2. **Project Data Analysis** – Analyzes physical, geographical, and financial data for accurate classification.
3. **Explainable AI** – Provides confidence scores and key influencing factors.
4. **Decision Support** – Assists planners and officials in transparent project monitoring and scheme allocation.
5. **Project Search & Insights** – Enables filtering by state, district, scheme, cost, and status.
6. **Interactive Dashboard** – Visualizes scheme distribution, project trends, budget utilization, and model performance.

Proposed Solution -AI-Powered PMGSY Scheme Classification System

The proposed solution is an AI-powered PMGSY Scheme Classification System built using IBM AutoAI, IBM watsonx.ai, and IBM Cloud Lite to automatically classify rural road and bridge projects into the correct PMGSY scheme.

Key Modules

Data Preprocessing – Cleans data, handles missing values, and prepares features using IBM AutoAI.

Scheme Classification – Predicts the correct PMGSY scheme (PMGSY-I, PMGSY-II, RCPLWEA, etc.) from project details.

Prediction & Decision Support – Provides scheme prediction with confidence scores for better decision-making.

Analytics Dashboard – Visualizes scheme distribution, project insights, and model performance.

Deployment & Evaluation

Real-Time Deployment – Model deployed on IBM watsonx.ai with a Streamlit web interface for instant predictions.

Performance Evaluation – Assessed using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to ensure reliable classification.

Technology Used

IBM AutoAI – Automated preprocessing, feature engineering, model training, and hyperparameter tuning.

IBM watsonx.ai – Deploys the trained model for real-time predictions.

IBM Cloud Lite – Secure and scalable cloud hosting environment.

Scikit-learn – Data preprocessing, feature transformation, and model evaluation.

XGBoost – Primary algorithm for accurate PMGSY scheme classification.

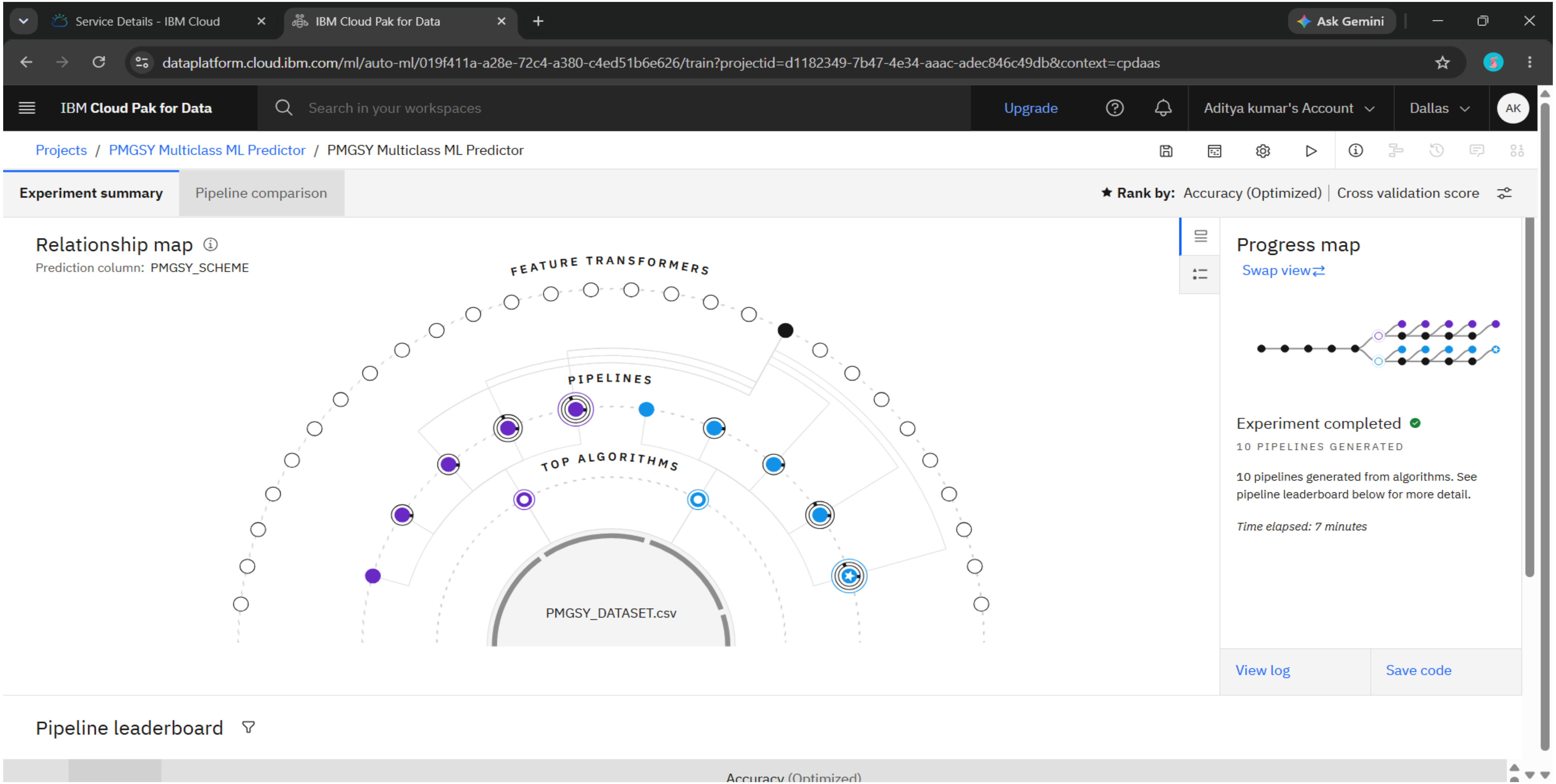
Pandas & NumPy – Data cleaning, preprocessing, and analysis.

Streamlit – Interactive web application for real-time predictions.

Matplotlib & Plotly – Visualization of project insights and model performance.

AI Kosh PMGSY Dataset – Historical dataset used for training, validation, and testing.

Result


Service Details - IBM Cloud
IBM Cloud Pak for Data
Ask Gemini
dataplatform.cloud.ibm.com/ml/auto-ml/019f411a-a28e-72c4-a380-c4ed51b6e626/train?projectid=d1182349-7b47-4e34-aaac-adec846c49db&context=cpdaas
IBM Cloud Pak for Data
Search in your workspaces
Upgrade
Aditya kumar's Account
Dallas
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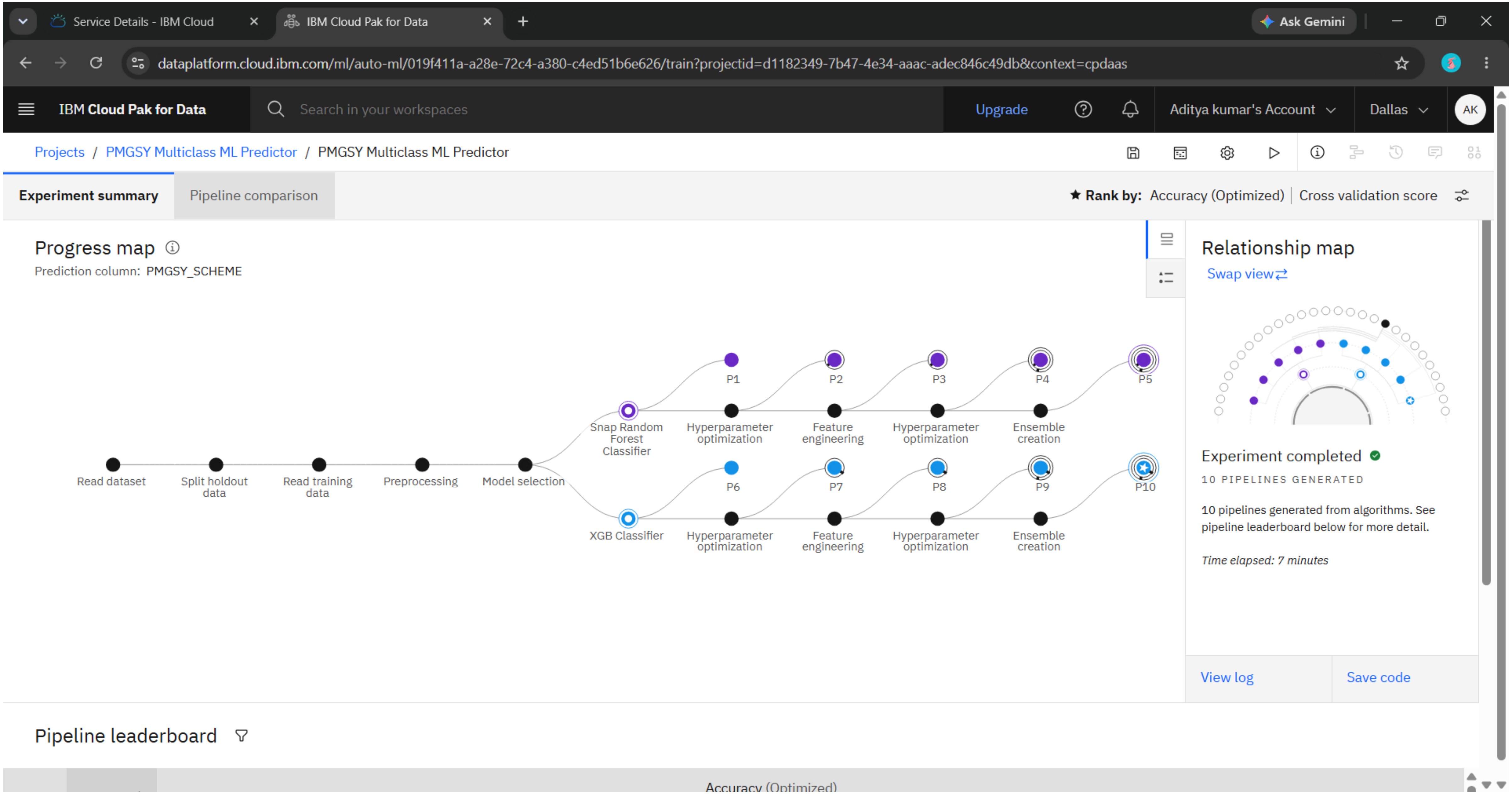
Projects / PMGSY Multiclass ML Predictor / PMGSY Multiclass ML Predictor
Experiment summary
Pipeline comparison
Rank by: Accuracy (Optimized)
Cross validation score
Relationship map
Prediction column: PMGSY_SCHEME
FEATURE TRANSFORMERS
PIPELINES
TOP ALGORITHMS
PMGSY_DATASET.csv
Progress map
Swap view
Experiment completed
10 PIPELINES GENERATED
10 pipelines generated from algorithms. See pipeline leaderboard below for more detail.
Time elapsed: 7 minutes
View log
Save code
Pipeline leaderboard
Accuracy (Optimized)

Result


Service Details - IBM Cloud
IBM Cloud Pak for Data
Ask Gemini
dataplatform.cloud.ibm.com/ml/auto-ml/019f411a-a28e-72c4-a380-c4ed51b6e626/train?projectid=d1182349-7b47-4e34-aaac-adec846c49db&context=cpdaas
IBM Cloud Pak for Data
Search in your workspaces
Upgrade
Aditya kumar's Account
Dallas
AK
Projects / PMGSY Multiclass ML Predictor / PMGSY Multiclass ML Predictor
Experiment summary
Pipeline comparison
Rank by: Accuracy (Optimized) | Cross validation score
Progress map
Prediction column: PMGSY_SCHEME
Read dataset
Split holdout data
Read training data
Preprocessing
Model selection
Snap Random Forest Classifier
XGB Classifier
Hyperparameter optimization
Feature engineering
Hyperparameter optimization
Ensemble creation
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
Relationship map
Swap view
Experiment completed
10 PIPELINES GENERATED
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Save code
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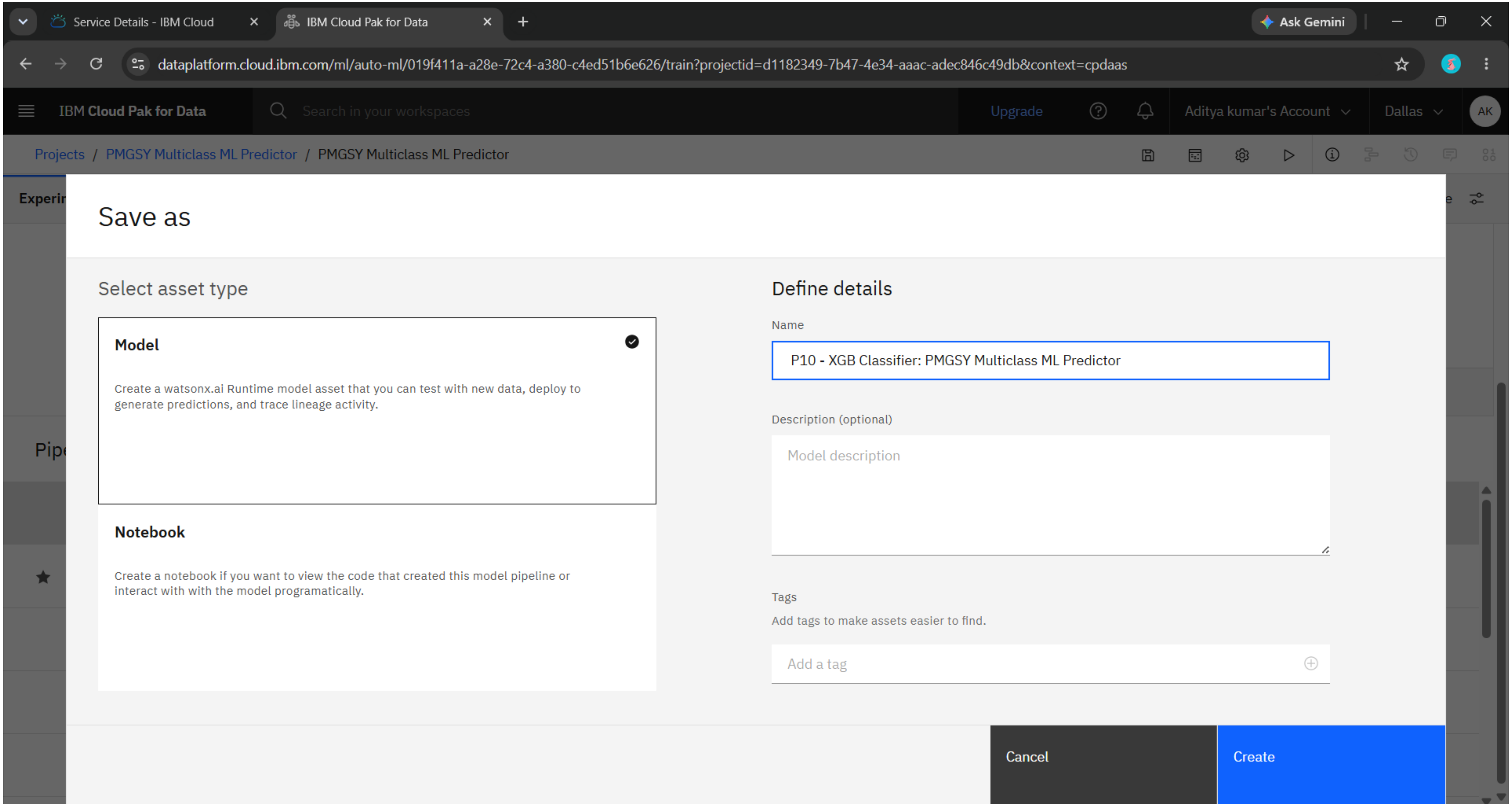
Accuracy (Optimized)

Result


Service Details - IBM Cloud
IBM Cloud Pak for Data
Ask Gemini
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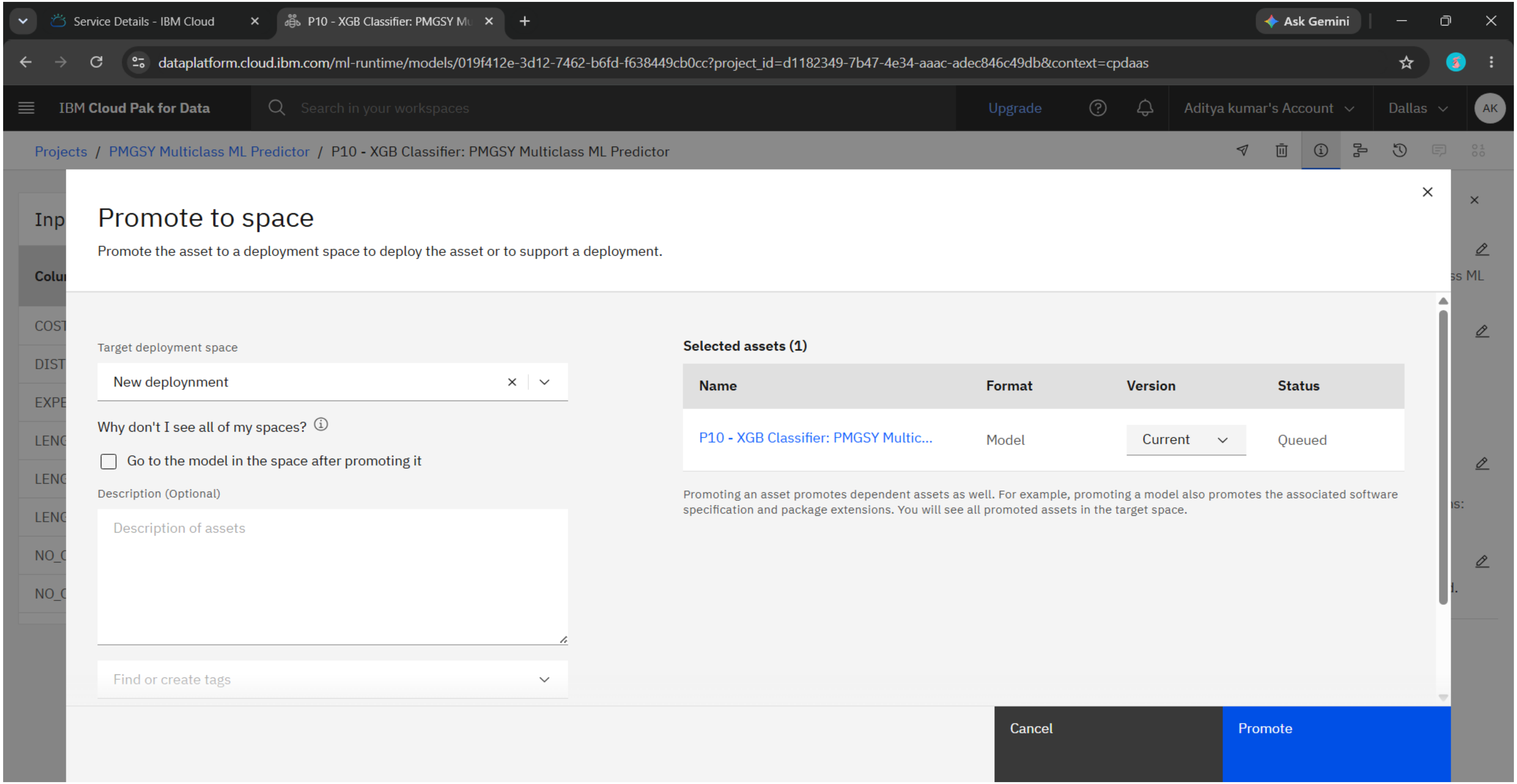
IBM Cloud Pak for Data
Search in your workspaces
Upgrade
Aditya kumar's Account
Dallas
AK
Projects / PMGSY Multiclass ML Predictor / PMGSY Multiclass ML Predictor
Save as
Select asset type
Model
Create a watsonx.ai Runtime model asset that you can test with new data, deploy to generate predictions, and trace lineage activity.
Notebook
Create a notebook if you want to view the code that created this model pipeline or interact with with the model programatically.
Define details
Name
P10 - XGB Classifier: PMGSY Multiclass ML Predictor
Description (optional)
Model description
Tags
Add tags to make assets easier to find.
Add a tag
Cancel
Create

Result


Service Details - IBM Cloud
P10 - XGB Classifier: PMGSY M
Ask Gemini
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IBM Cloud Pak for Data
Search in your workspaces
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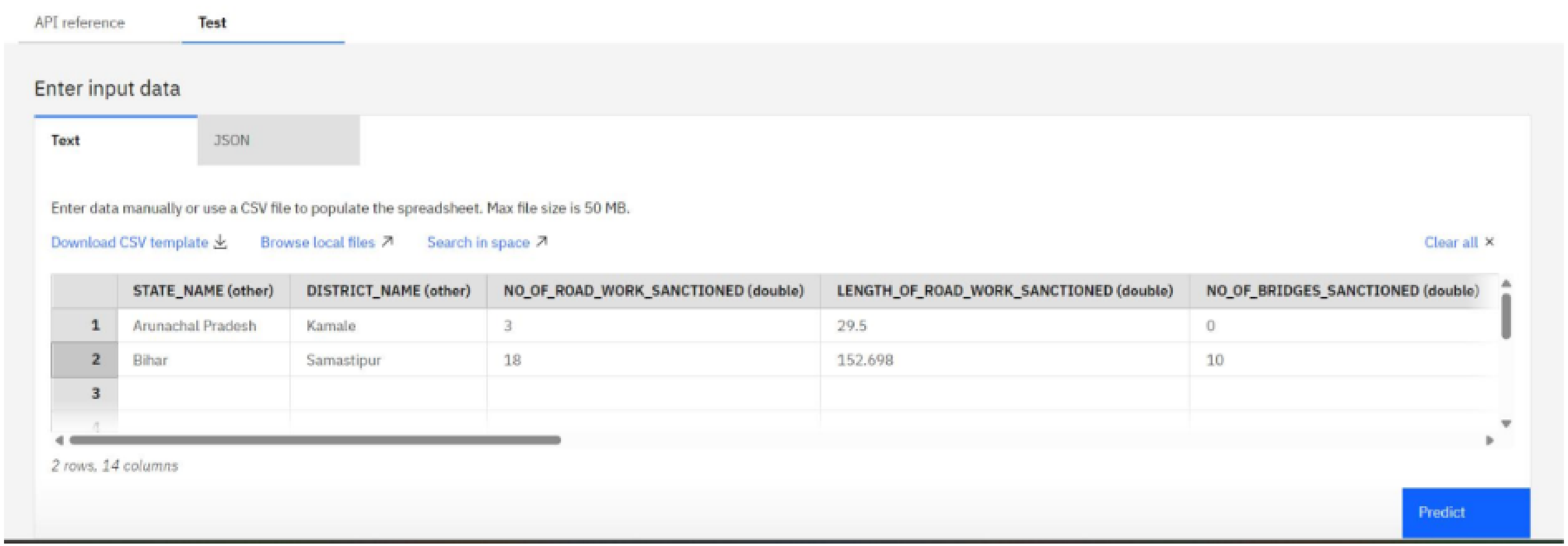
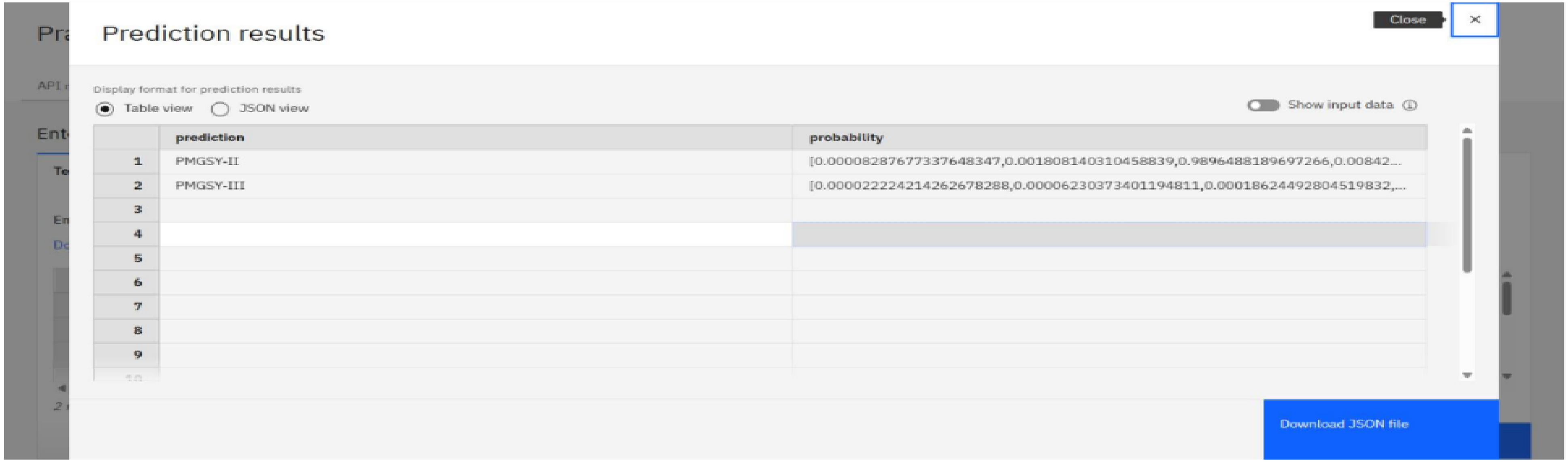
AK
Projects / PMGSY Multiclass ML Predictor / P10 - XGB Classifier: PMGSY Multiclass ML Predictor
Promote to space
Promote the asset to a deployment space to deploy the asset or to support a deployment.
Target deployment space
New deployment
Why don't I see all of my spaces?
Go to the model in the space after promoting it
Description (Optional)
Description of assets
Find or create tags
Selected assets (1)
Name
Format
Version
Status
P10 - XGB Classifier: PMGSY Multic...
Model
Current
Queued
Promoting an asset promotes dependent assets as well. For example, promoting a model also promotes the associated software specification and package extensions. You will see all promoted assets in the target space.
Cancel
Promote

Result

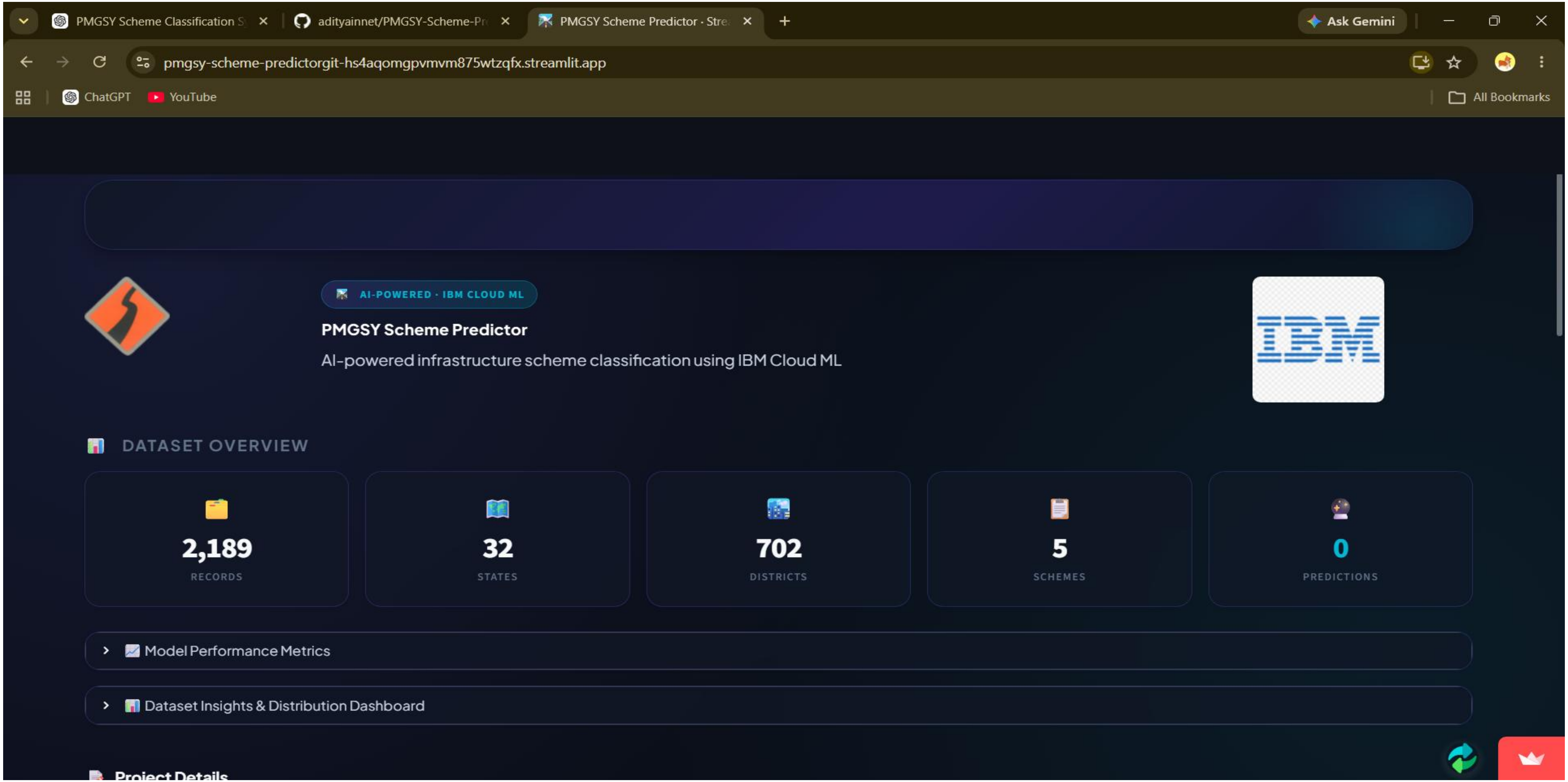


API reference
Test
Enter input data
Text
JSON
Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.
Download CSV template
Browse local files
Search in space
Clear all
STATE_NAME (other)
DISTRICT_NAME (other)
NO_OF_ROAD_WORK_SANCTIONED (double)
LENGTH_OF_ROAD_WORK_SANCTIONED (double)
NO_OF_BRIDGES_SANCTIONED (double)
1 Arunachal Pradesh Kamale 3 29.5 0
2 Bihar Samastipur 18 152.698 10
3
2 rows, 14 columns
Predict



Prediction results
Close
Display format for prediction results
Table view
JSON view
Show input data
prediction
probability
1 PMGSY-II [0.00008287677337648347,0.001808140310458839,0.9896488189697266,0.00842...
2 PMGSY-III [0.000022224214262678288,0.00006230373401194811,0.00018624492804519832,...
3
4
5
6
7
8
9
Download JSON file

Frontend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PMGSY Scheme Predictor · Stre
Ask Gemini
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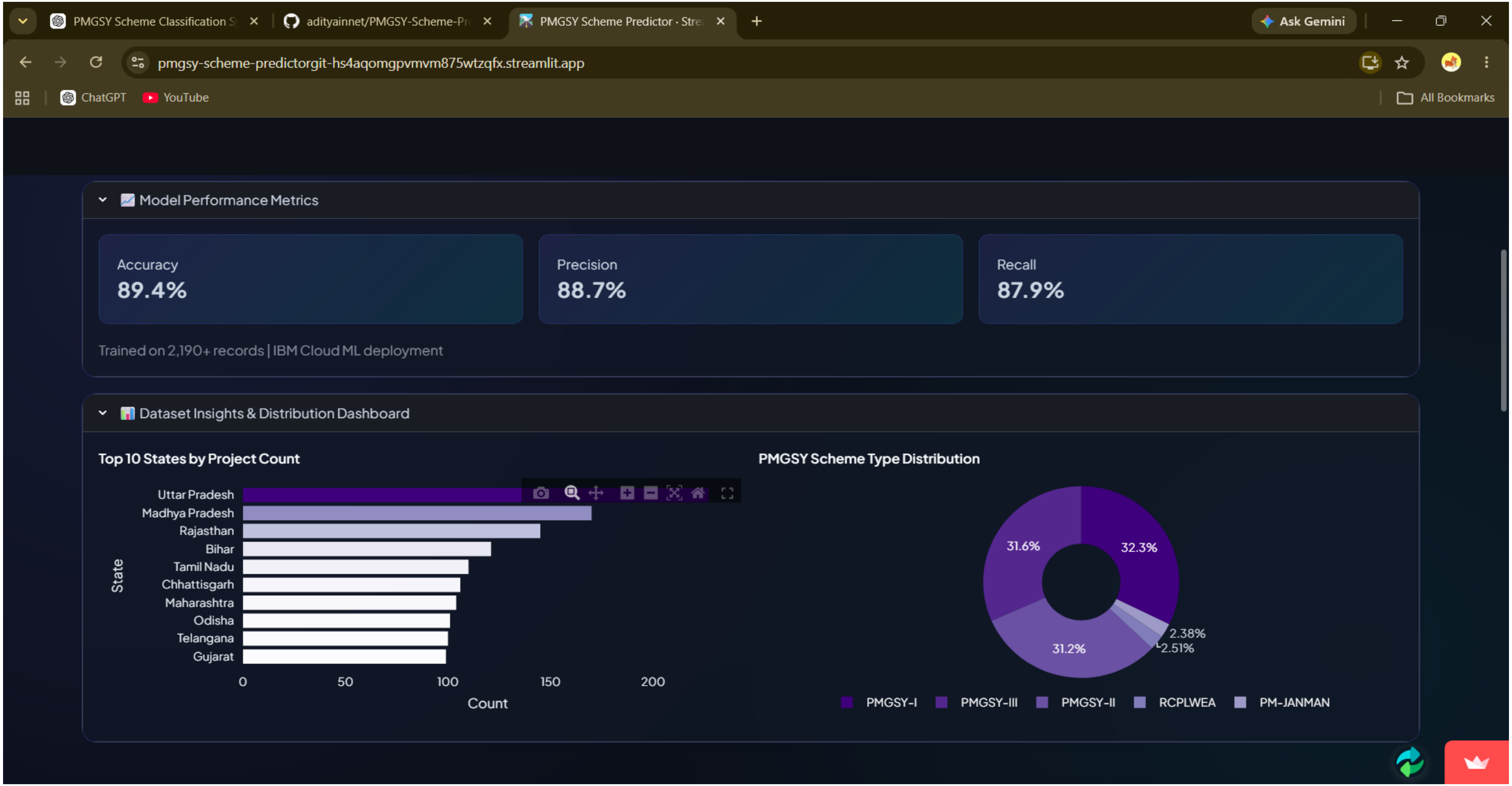
ChatGPT
YouTube
All Bookmarks
AI-POWERED · IBM CLOUD ML
PMGSY Scheme Predictor
AI-powered infrastructure scheme classification using IBM Cloud ML
IBM
DATASET OVERVIEW
2,189
RECORDS
32
STATES
702
DISTRICTS
5
SCHEMES
0
PREDICTIONS
Model Performance Metrics
Dataset Insights & Distribution Dashboard
Project Details

Frontend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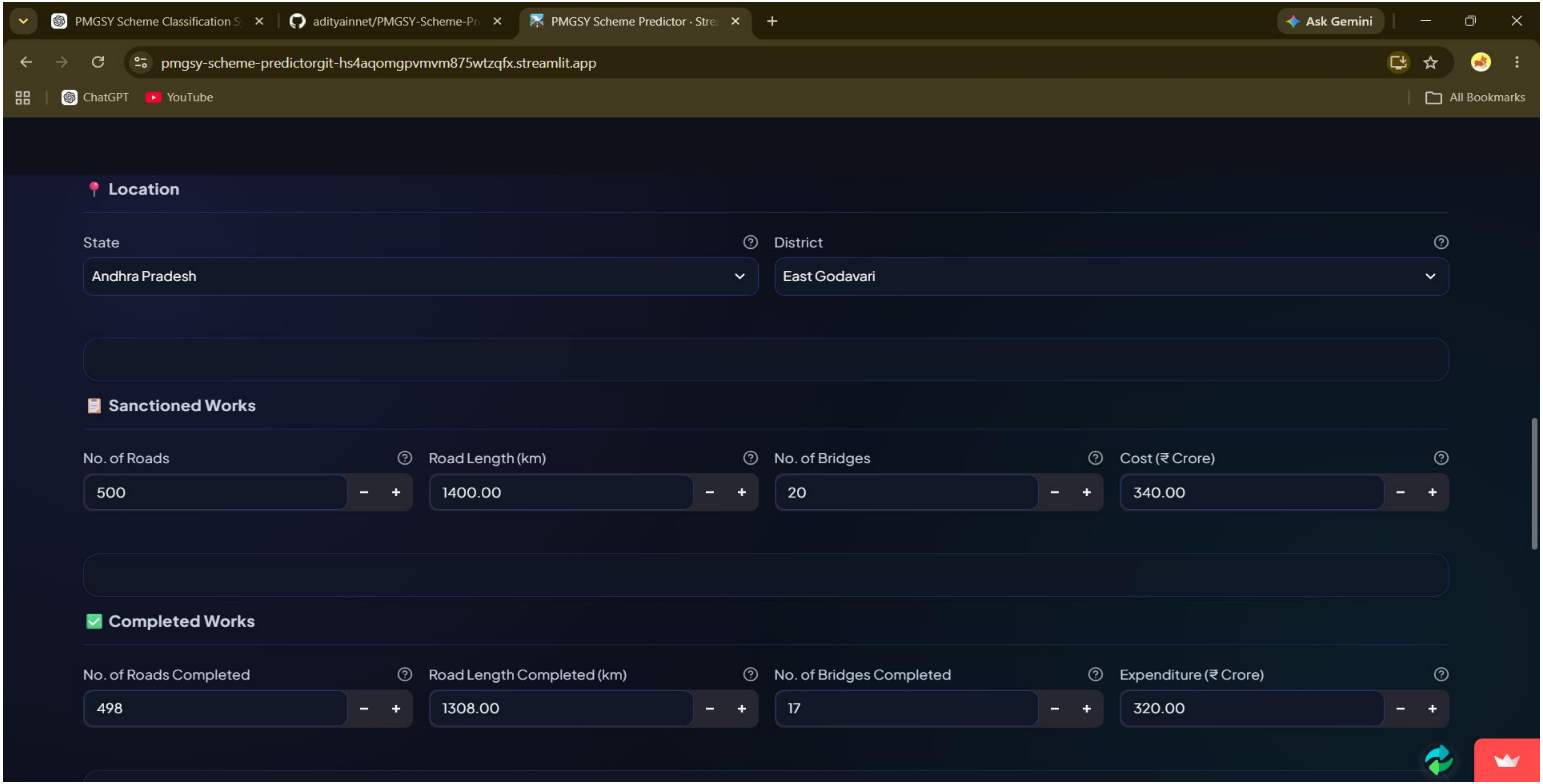
PMGSY Scheme Predictor · Stre
Ask Gemini
pmgsy-scheme-predictorgit-hs4aqomgpvmvm875wtzqfx.streamlit.app
ChatGPT
YouTube
All Bookmarks
Model Performance Metrics
Accuracy
89.4%
Precision
88.7%
Recall
87.9%
Trained on 2,190+ records | IBM Cloud ML deployment
Dataset Insights & Distribution Dashboard
Top 10 States by Project Count
Uttar Pradesh
Madhya Pradesh
Rajasthan
Bihar
Tamil Nadu
Chhattisgarh
Maharashtra
Odisha
Telangana
Gujarat
State
0
50
100
150
200
Count
PMGSY Scheme Type Distribution
31.6%
32.3%
2.38%
2.51%
31.2%
PMGSY-I
PMGSY-III
PMGSY-II
RCPLWEA
PM-JANMAN

Frontend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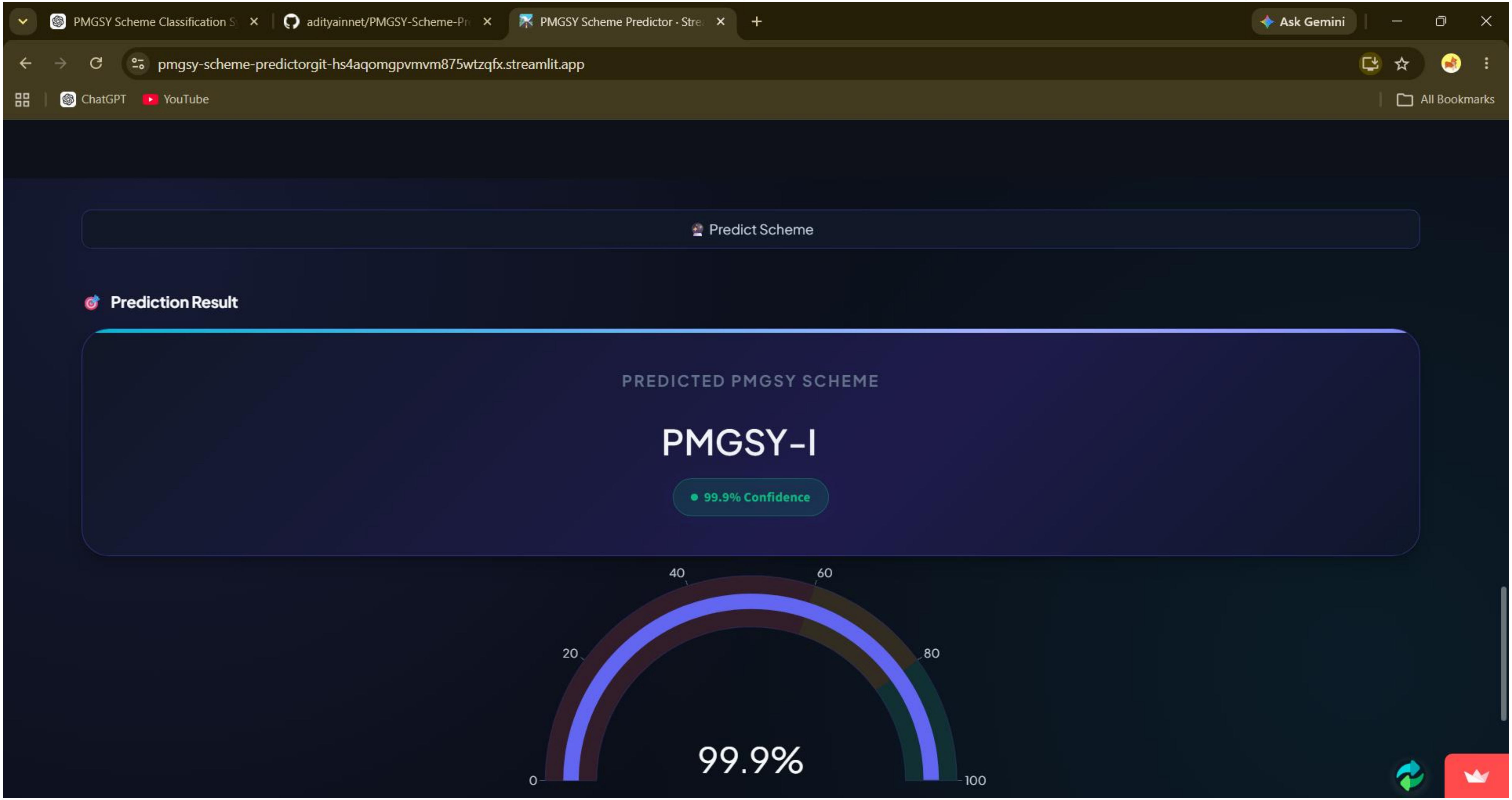
ChatGPT
YouTube
All Bookmarks
Location
State
Andhra Pradesh
District
East Godavari
Sanctioned Works
No. of Roads
500
Road Length (km)
1400.00
No. of Bridges
20
Cost (₹ Crore)
340.00
Completed Works
No. of Roads Completed
498
Road Length Completed (km)
1308.00
No. of Bridges Completed
17
Expenditure (₹ Crore)
320.00

Frontend


PMGSY Scheme Classification S
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PMGSY Scheme Predictor · Stre
Ask Gemini
pmgsy-scheme-predictorgit-hs4aqomgpvmvm875wtzqfx.streamlit.app
ChatGPT
YouTube
All Bookmarks
Predict Scheme
Prediction Result
PREDICTED PMGSY SCHEME
PMGSY-I
99.9% Confidence
0
20
40
60
80
100
99.9%

Novelty and Uniqueness

Key Features of the Proposed PMGSY Scheme Classification System:-

Automated Scheme Classification – Accurately predicts the appropriate PMGSY scheme, reducing manual effort.
IBM AutoAI Optimization – Automates preprocessing, feature engineering, model selection, and hyperparameter tuning.
High Prediction Accuracy – Uses advanced ML models (e.g., XGBoost) for reliable scheme classification.
Real-Time Prediction – Instantly classifies new road and bridge projects through a web interface.
Explainable AI – Displays prediction confidence and key influencing factors for transparency.
Interactive Dashboard – Visualizes scheme distribution, project insights, and model performance.
Scalable & Cloud-Ready – Built on IBM watsonx.ai and IBM Cloud Lite for secure and scalable deployment.

Git Hub Link

Github URL –https://github.com/adityainnet/PMGSY-Scheme-Predictor.git

Streamlit Deployment Link:-https://pmgsy-scheme-predictorgit-hs4aqomgpvmvm875wtzqfx.streamlit.app/

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Future Scope

1. Real-Time Data Integration
Integrate live PMGSY project data from field offices and government portals to enable real-time scheme classification and continuous project monitoring.
2. Interactive Performance Dashboard
Develop an advanced dashboard to visualize scheme-wise project distribution, project status, regional insights, budget utilization, and model performance.
3. Predictive Analytics
Extend the solution to predict project delays, cost overruns, funding requirements, and project completion timelines using historical data.
4. Explainable & Advanced AI
Incorporate Explainable AI (XAI) to justify predictions and use Natural Language Processing (NLP) to analyze project reports, inspection notes, and official documents for more accurate classification and decision support.

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Certificates Earned:

In recognition of the commitment to achieve professional excellence

Getting Started with Artificial Intelligence

Aditya Kumar

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Jun 18, 2026
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/d6b0c6bd-50f0-4cfb-bfce-3919adfe8549

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Certificates Earned:

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Aditya Kumar

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 18 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.